Suchetan Mukherjee

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Experience

Highradius Technologies — Software Development Engineer 1 *July 2025 – Present*

- Designed and built **RESTful microservice APIs** in **Java/Spring Boot** with **JSON/DTO contracts**, documented via Swagger/OpenAPI and consumed across Supplier Portal, Onboarding, and ERP services.
- Architected event-driven microservice communication using **Kafka**, building producer/consumer workflows that ingest and process **1M+ invoices** from Accounts Payable, with **idempotent consumers** ensuring exactly-once processing and near real-time status propagation.
- Built a full-stack **Supplier Onboarding Form Builder** with **React/TypeScript and Spring Boot**, featuring configurable sections, dynamic validations, and a **tenant-level EAV architecture**; the product has **onboarded 5 new clients** since launch.
- Orchestrated supplier lifecycle workflows using **Flowable/GWEN**, triggering API and inter-service notifications with **role-based authorization** across Supplier Master, Onboarding, and Admin modules.
- Engineered **ERP posting pipelines** integrating **Snowflake** data through object/data mapping layers, applying **database indexing** and **caching** to improve query performance, transferring CSV payloads via **AWS S3**.
- Integrated **AI developer tools** (GitHub Copilot, Claude Code) into daily development and debugging workflows, and configured **custom AI agents** using instruction and skill markdown files with tool/MCP integrations.

Highradius Technologies — Product Engineering Intern *May 2024 – June 2025*

- Developed automated transaction-tagging pipelines using **Spring Boot, SQL, and Snowflake** for backend data retrieval, processing, and aggregation, powering automated client cash flow categorization.
- Implemented production observability using the **EFK Stack and Grafana**, monitoring API failures, success rates, **P95/P99 latency**, and JVM metrics to identify efficiency and performance improvements.
- Increased backend test coverage from 0 to **80%** by writing comprehensive **JUnit and integration test suites**.
- Resolved **50+ production issues per sprint** as part of the SWAT team, including **7 critical incidents and 4 blocker-severity pages**, performing root-cause analysis and closing criticals same-day, blockers within the hour.
- Developed **Python automation scripts** for transaction analytics, dynamically executing SQL workflows and generating WBR-ready reports for client insights.

Samsung R&D — Research Intern, Prism Program *March 2023 – March 2024*

- Benchmarked and fine-tuned **10+ SOTA computer vision segmentation models** — including **SAM, SegFormer, OneFormer, and DINOv2** — using **PyTorch/HuggingFace** across COCO, Cityscapes, ADE20K, and DIS5K datasets.
- Custom-trained Meta's **Segment Anything Model (SAM)** on DIS5K and built evaluation pipelines measuring segmentation accuracy, inference speed, and model robustness.

Projects

LifeLink — Blood Donor–Hospital Matching Platform | Live Demo | Source *2026*

- Built and deployed a full-stack donor-matching platform with **Spring Boot 3, PostgreSQL, Redis, and React**, exposing secured **REST APIs** for four distinct roles via **JWT access/refresh token authentication** with rotation and BCrypt hashing.
- Designed a **Redis GEO-indexed matching engine** resolving blood-group compatibility and ranking donors by distance, backed by a **Spring State Machine** request lifecycle with guard conditions and a full audit trail.
- Implemented **asynchronous RabbitMQ notification pipelines** with dead-letter queues and retry backoff, plus **Quartz-scheduled jobs** for escalation, expiry, and eligibility refresh.
- Applied **Redis-backed rate limiting** and response caching on hot endpoints, with **Flyway**-versioned migrations and Dockerized dependencies.

Technical Skills

- **Languages:** Java, Python, JavaScript, TypeScript, SQL
- **Frameworks & Libraries:** Spring Boot, Spring Security, React.js, JUnit, Mockito, Flowable, PyTorch, Pandas
- **APIs & Architecture:** REST APIs, JSON/DTO Modeling, Swagger/OpenAPI, Microservices, Apache Kafka, RabbitMQ, Event-Driven Architecture, Asynchronous Processing, JWT/RBAC, Idempotency, Caching, Rate Limiting
- **Databases & Data Platforms:** PostgreSQL, MySQL, Snowflake, Redis, Database Indexing, Flyway
- **Cloud, DevOps & Observability:** AWS S3, Docker, ArgoCD, Jenkins, CI/CD, EFK Stack, Grafana, JVM Monitoring
- **AI & ML:** PyTorch, HuggingFace, Model Fine-Tuning, GitHub Copilot, Claude Code, AI Agent Development, MCP
- **Tools & Practices:** Git, GitHub, Maven, Gradle, Postman, Agile/Scrum, Unit & Integration Testing

Achievements & Publications

- **Journal Publication:** "Early ASD Screening in Toddlers: A Comprehensive Stacked ML Approach," Intl. Journal of Computing & Digital Systems — achieved **99.14% accuracy**.
- **Book Chapter:** "Digital Twins in the Metaverse," a 20+ page chapter in *Healthcare Services in the Metaverse*.
- **Research Impact:** Combined publications have garnered **500+ reads and 8 citations**.

Education

Bachelor of Technology in Computer Science Engineering *2021 – 2025*
Kalinga Institute of Industrial Technology, Bhubaneswar *CGPA: 8.81*

Indian School Certificate (12th) — Calcutta Boys' School, Kolkata *2021 | 96.5%*

Indian Certificate of Secondary Education (10th) — Calcutta Boys' School, Kolkata *2019 | 95.2%*